

Computer Space

Score Digit Extension

(Interposer PCB for Revision 1 Sync-Boards)
Version 0.95

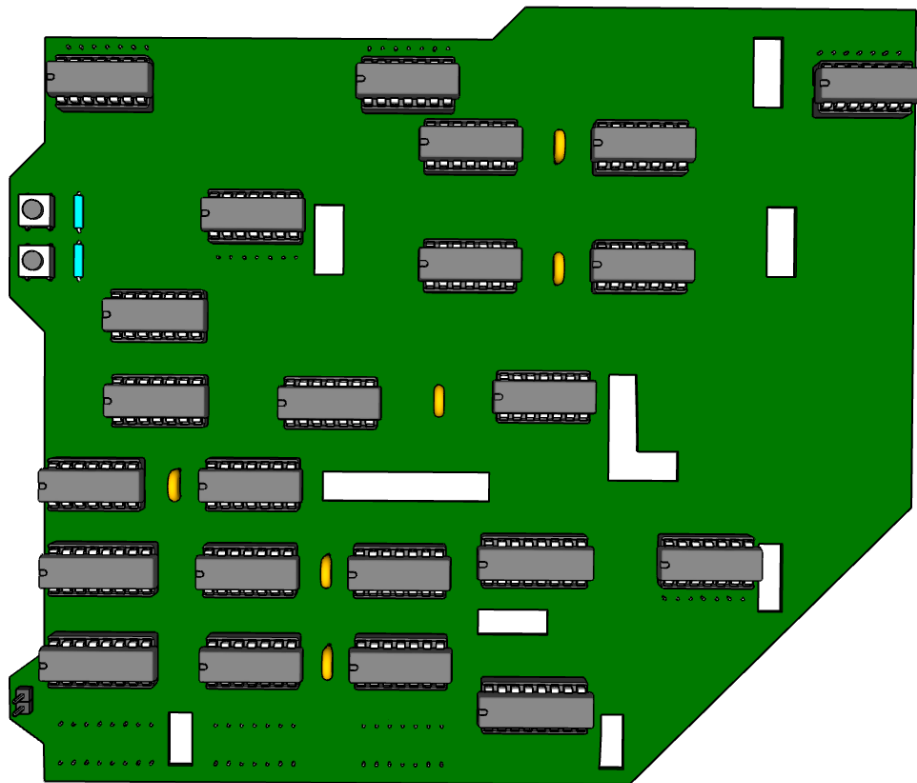


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Preface

Most likely everyone that somehow deals with the topic of the single player Computer Space cabinet over a longer period of time will come across the fact, that the score counting is kind of unique, interesting and unusual. Once a score exceeds 9 it will either revolve to 0 (saucer score) or result in an unrecognizable hieroglyph (rocket score).

A design flaw (really? more on that later) that exists as long as the game itself, that was never resolved within over 55 years and somehow become an iconic 'feature' of the first commercial video game.

Unfortunately, that's the kind of rapid hole I cannot resist!

Once I accepted the challenge, I didn't care if it makes much sense or being economical. I just enjoyed the fun of the challenge and as it resulted in a feasible solution, I'm happy to share it with the community.

In case you decide to follow into the rapid hole, have fun with the build and happy arcade gaming!

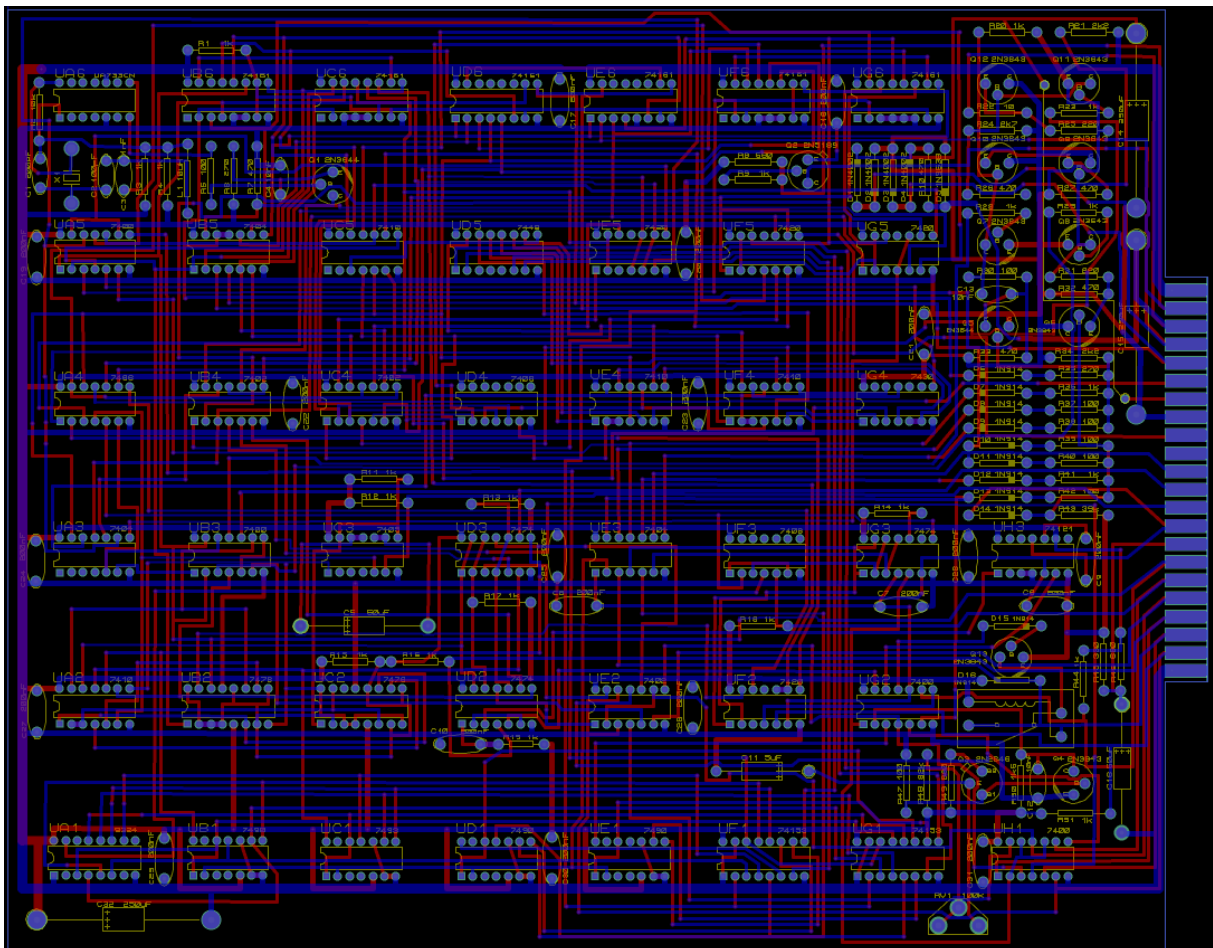
Best regards,
Stefan Burger

Introduction

Design Issue?

It is a bit unfair to talk about a 'design issue' as in my opinion it was most likely an educated decision to implement an acceptable compromise for the score counting. My lessons learned from dealing with the issue is, that it is and was back then not easy at all to implement a second digit for the scores. It would have required not only additional counter ICs but as well a lot more components for additional logic.

Having a look on the Gerber-files of the revisions 1 Sync-board (kind of an X-ray of the whole thing), it becomes clear, that there was just no sufficient space left to place all necessary components as well as necessary trace routing. It would have required either a 4th board or a significant increase of the PCB's footprint and of course result in significant additional costs for production. It is also worth noticing, that the footprint of all 3 boards should be kept similar as they should end up in the very same board cage.



Considering a limited game time, it might have seemed acceptable to stay with one-digit scores.

An interesting observation in this context is the fact that normally 74161 timers have been used as hexadecimal counter (0-15). The only counters that deviating are those for scores and game timers.

In the area of score and game timers, the 7490 dominates, which are BCD (binary decimal counters). In that area, the rocket score, using a 7493 counter is one more exception. The 7493 is a binary counter that supports counting from 0 to 15 before it revolves to 0. Why not using a 7490 BCD as for the other score/ timer counters or a 74161 hex-counter as typically used for other purposes?

Well, I'm going to assumptions here. I think it was a late decision to replace the rocket score counter from a 7490 (0-9) with a 7493 (0-15) as they are pin compatible and make a cheap and easy replacement without the need to reroute all traces (back then without CAD software quite a challenge). This decision was most likely driven by the consideration that players might get frustrated reaching a higher score then the saucers and would not get rewarded with an extended play in "Hyperspace", just because of technical shortcomings.

Extending the counter from a maximum count from 9 to 15 might seemed to be an acceptable compromise, considering the game time limit.

Talking about game time limit: As this interposer board will allow score counting for saucer and rockets up to 99, it might be worth considering an extension of game time for longer game sessions. This can be already adjusted to a certain degree with the 100k potentiometer at RV1. The game time could be extended even further by replacing it with a potentiometer of much higher maximum resistance (e.g. 500k Ohm).

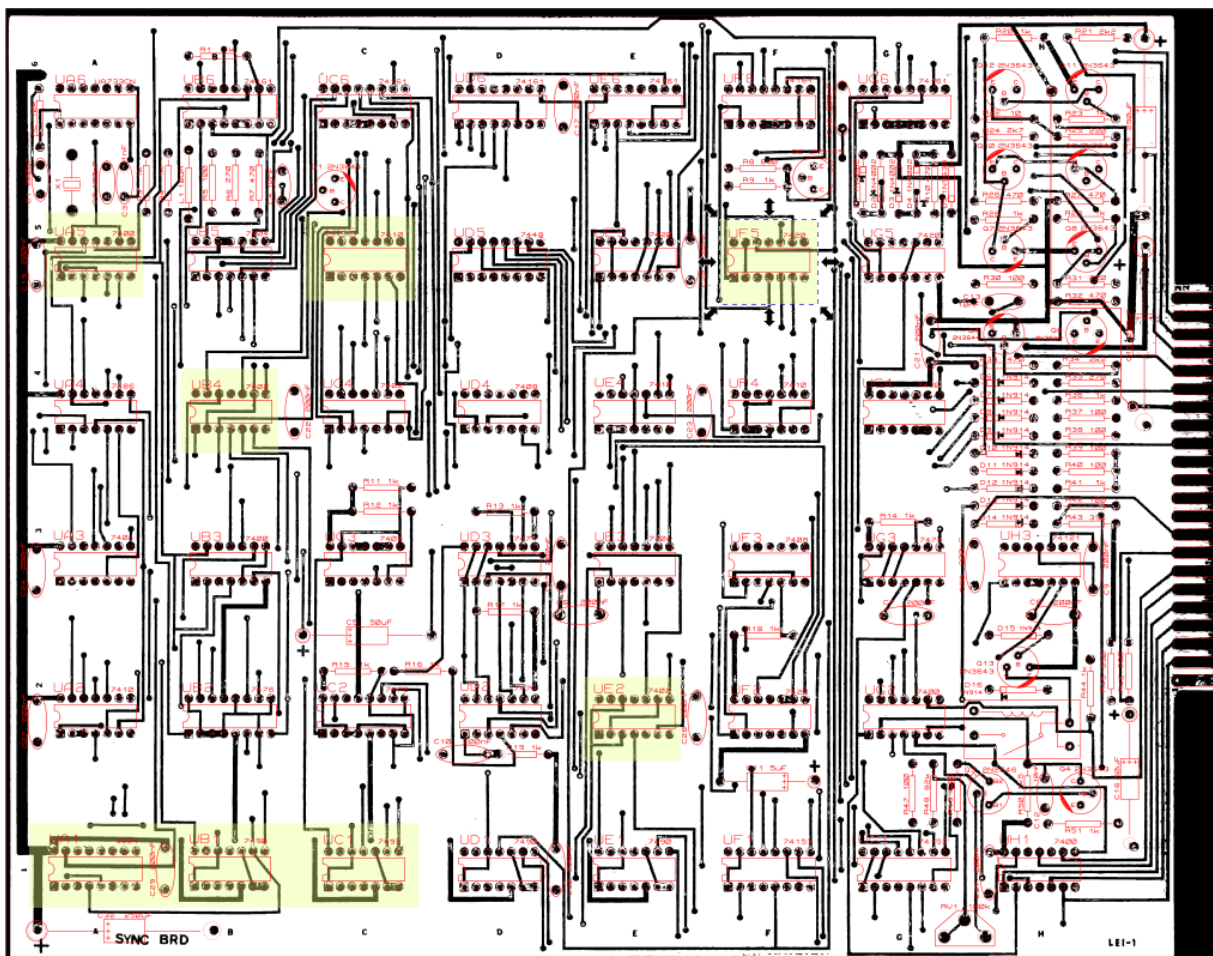
Compatibility

The Digit-Extension interposer board is compatible with the revision 1 Sync-board and works with the original Nutting build boards from the early 70's as well with replica boards, based on GitHub user pong74ls Gerber-files.

The revision 2 Sync-boards are not compatible as they seem to follow the same logic but have the positioning of the ICs on the Sync-board PCB at different locations (more on that later).

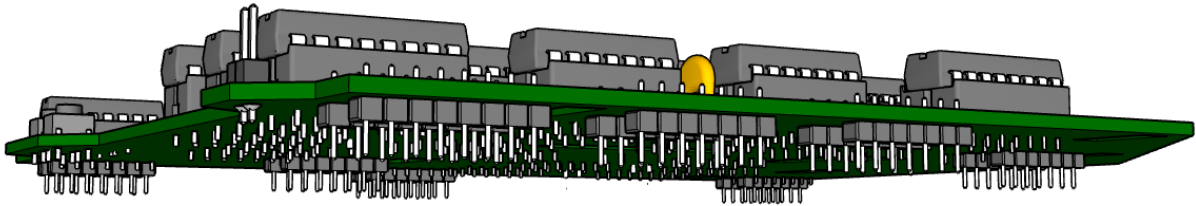
GitHub user pong74ls did an excellent job with the replication not only in terms of circuit logic but in terms of physical dimensions of the PCB as well. I checked the placement of the components and they are all where they are on the original Nutting boards, with an incredible precision.

The following image was used to check the precise alignment. All black traces and markings are the result of a scan of an unpopulated original Nutting Sync-board, overlaid with the replica layout in red color. The yellow highlighted ICs are relevant for the compatibility of the interposer PCB. The placement is precise, down to tenth of a millimeter:



Preconditions

The general concept how the interposer board interacts with the Sync-board is by retrieving necessary information, extracted from IC sockets at the Sync-board and feedback enhanced signaling to it via pin-head connectors that will be placed into IC sockets at the Sync-board.



Besides of socketing eight ICs on the revision 1 Sync-board, no additional changes are necessary. This kind of “minimally invasive” modification will allow for a fast and easy reversion to the original setup, whenever needed.

Sync-board revision 1 only

Based on available documentation, there are at least two revisions for the single player Sync-board. While the general logic of both revisions seems to be identical, the change of IC names indicates a different physical location of the ICs on the board (most likely with the intention to optimize the routing of copper traces).

As a result, an adapted version of the interposer board for the 2nd revision Sync-board would be theoretically possible, but would require a complete realignment and positioning of all components on the interposer board. Unfortunately, there is no sufficient documentation regarding the 2nd revision of the Sync-board available, that would allow the exact positioning, especially for the pin-head connectors that require a precise alignment to make contact to the Sync-board’s IC-sockets.

How to determine the revision of the Sync-board

Without access to documentation or images of the PCB Layout for the 2nd revision this question is in fact not easy to answer. However, there is documentation for both revisions regarding the schematics that carrying the names of ICs. The name convention seems to be based on the physical location of the IC on the board, so that the second and third digit are pointing to a coordinate on the PCB. The coordinates on the horizontal are labeled from A to H while on the vertical are counting from 1 to 6. For example, UA1 is the IC on the lower left side, where as UG6 is the IC on the upper right side (with the Sync-boards edge connector oriented to the right).

Based on this assumption and with reference to the schematic sheets, it becomes possible to detect deviations of the IC placement. Unfortunately, there are reports that ICs were made unrecognizable by covering their markings with paint (most likely with the intention to protect intellectual property).

However, there are differences in the number of pins (DIP-14 versus DIP-16). Some counting of IC-pins in certain locations might help to identify the revision of the Sync-board.

		A		B		C		D		E		F		G		H		I		J	
6	v1	UA733CN	14	74161	16	74161	16	74161	16	74161	16	74161	16	74161	16		0		0		0
	v2	7400	14	7402	14	74121	14	7420	14		0		0		0		0		0	UA733CN	14
5	v1	7400	14	7404	14	7410	14	7448	16	7420	14	7420	14	7420	14		0		0		0
	v2	7474	14	7474	14	7400	14	7474	14	7404	14		0		0		0		0	7400	14
4	v1	7486	14	7402	14	7402	14	7408	14	7410	14	7410	14	7430	14		0		0		0
	v2	7410	14	7476	16	7476	16	7405	14		0		0		0		0		0	7430	14
3	v1	7404	14	7400	14	7405	14	7474	14	7404	14	7408	14	7474	14	74121	14		0		0
	v2	74153	16	7490	14	7493	14	7490	14	7490	14	74153	16	7448	16	7410	14		0	7410	14
2	v1	7410	14	7476	16	7476	16	7474	14	7402	14	7420	14	7400	14		0		0		0
	v2	9324	16	7404	14	7408	14	7402	14	7420	14	7420	14	7420	14	7402	14		0	7410	14
1	v1	9324	16	7490	14	7493	14	7490	14	7490	14	74153	16	74153	16	7400	14		0		0
	v2	74161	16	74161	16	74161	16	74161	16	74161	16	74161	16	7404	14	7400	14		0	7486	14

As above table illustrates, UB1 (first row, starting on the lower side; second column, starting from the left) is in revision 1 a 7490 with 14 pins, while it seems to be in revision 2 a 74161 with 16 pins. The table provides an overview, based on deviations in the naming of the IC and the assumption, that the name indicates its position on the Sync-board.

Sockets for eight ICs on the Sync-board

Here is a short overview on which ICs on the Sync-board have to be socketed and why:

Name	Type	Reasoning
UA1	9324	The 9324 is a 5-bit comparator that is used to compare the scores for the rocket and the saucer. In case the extended play feature is enabled and the rocket score is higher than the saucer score, the game will continue in "Hyperspace" for another 100 game time units. As both timers will be extended to 8 bits, the comparison has to be enhanced accordingly. This requires a second 9324 as well as additional and-/or-gates. If the timer reaches it's maximum, the tenth of each score counter will be compared, as well as the units of both score counters.
UB1	7490	This is the 4-bit decimal counter for the saucer score. It will be chained with a second 7490 so that if it revolves from 9 to 0, it will trigger the counting of the tenth counter via a carry-over signal.
UC1	7493	This is the 4-bit hexadecimal counter for the rocket score. This will be replaced with a 7490 decimal counter and as well trigger a second 7490 for the tenth, just like it is done with the saucer score (see above).
UE2	7402	This is a NOR-gate that is used for collision detection. This is quite helpful as it allows to inject simulated collisions for easier testing of the interposer board.
UB4	7402	While the initial purpose of this IC is not relevant, it carries 3 of 4 bits of the upper half of the vertical sync-counter. The 4 most significant bits of the vertical counter are important as input to determine, whether the actual raster position is within a 16x16 pixel block on the screen, where one of the counters has to be displayed. This IC has been chosen instead of the vertical sync counter IC, because of its placement on the board. This allows to keep the footprint of the interposer board much smaller.
UA5	7400	This IC carries information about the actual horizontal raster position. This information is important as input to determine, whether the actual raster position is within a 16x16 pixel block on the screen, where one of the counters has to be displayed.
UC5	7410	This IC generates a signal that enables or disables the output of a digit on the screen, so that digits are only visualized at the relevant screen regions. As the original logic suppresses output at locations where the additional digits have to be displayed, the control signal has to be generated by own logic and the output has to be injected back into the Sync-board.
UF5	7420	This IC carries the 4 th missing bit of the vertical counter and therefore completes the necessary information. For more details see UB4.

BOM

The Bill of Materials is as follows:

Category	Name	Type	Amount
IC-Sockets	(Interposer PCB)	DIP-14	19
	(Interposer PCB)	DIP-16	4
	(Sync-board)	DIP-14	7
	(Sync-board)	DIP-16	1
IC's	UA1A; UA1B	9324	2
	UB1A; UB1B; UC1A; UC1B	7490	4
	U1; U2	74153	2
	UE2	7402	1
	U4; U10; U11; U7	7432	4
	U9; U8	7408	2
	U3; U6	7421	2
	U12; U5	7404	2
	UB4	7402	1
	UA5	7400	1
	UC5	7410	1
	UF5	7420	1
Pin-Headers	(see Recommendations)	1x7, 2.54 pitch	14
	(see Recommendations)	1x8, 2.54 pitch	2
		1x2, 2.54 pitch	1
Resistors	R1; R2	1k-Ohm	2
Capacitors	C1; C2; C3; C4; C5; C6	220 nF	6
Tactile Switches	SW-RKT; SW-SRC	C&K PTS645SH50-2 LFS	2

The Gerber-files for producing the interposer PCB can be found at:

<https://github.com/SteBuTOS/ComputerSpaceDesign/tree/master/Add-ons/CS-Digits>

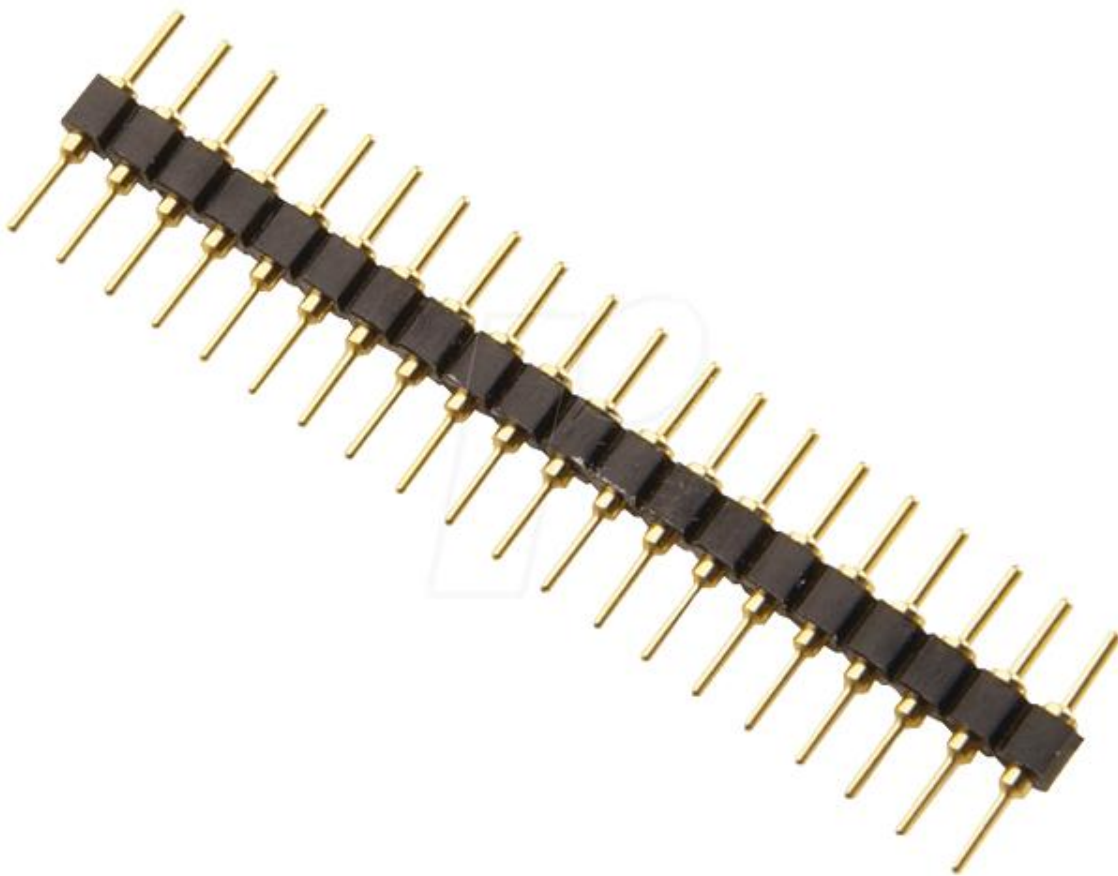
Recommendations

Pin-header recommendation

There are specific types of pin-headers available with optimized properties for the purpose to be used with IC-sockets. Their pins are much thinner and smoother and require less force to be applied.

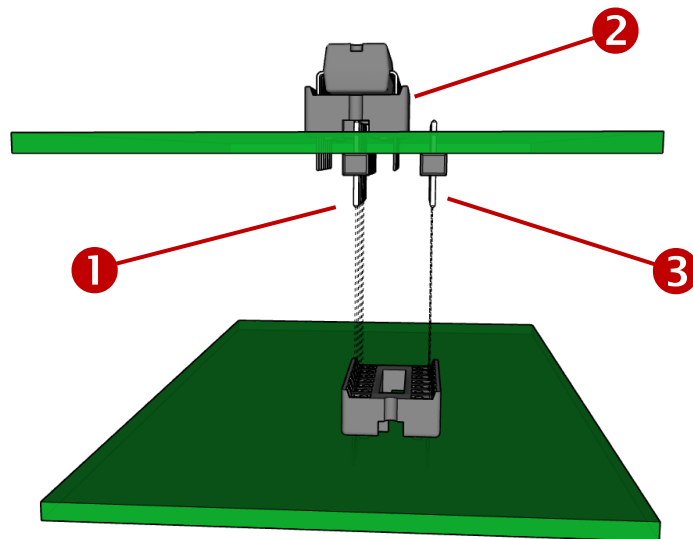
The eight relevant ICs have a total of 114 pins. The potential total friction by using less optimal pins would be quite high.

See the following image for reference:



Order of soldering

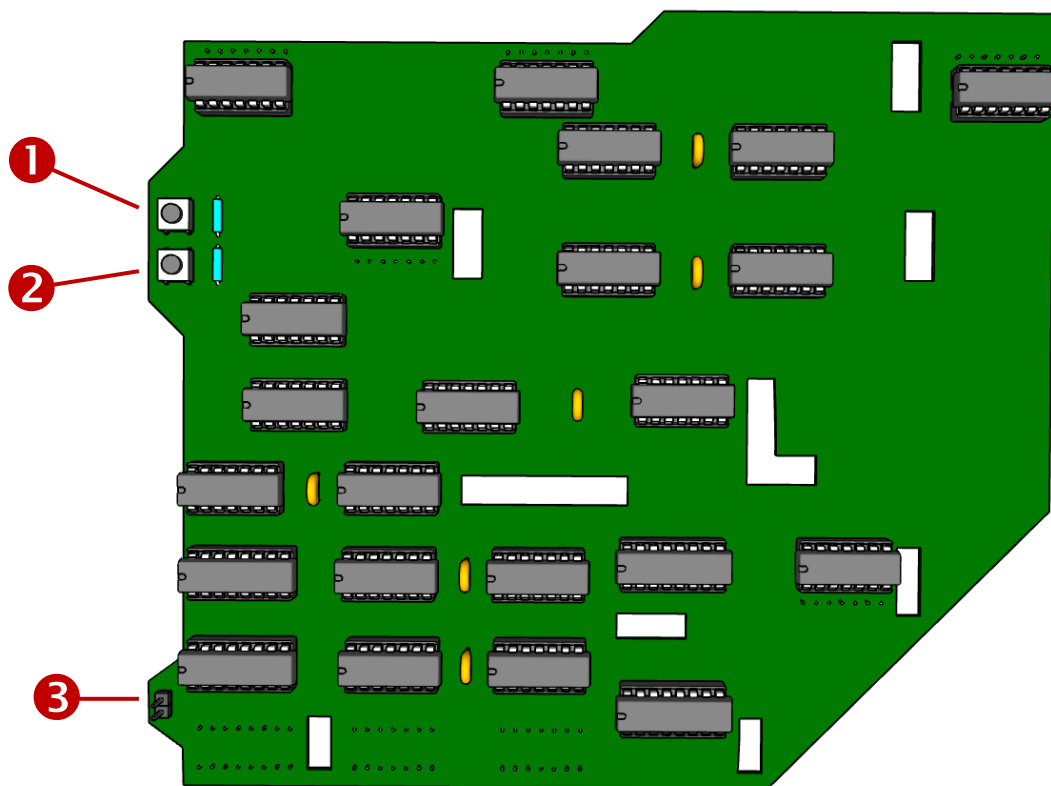
With the intention to keep the footprint of the interposer board as small as possible, some IC sockets on the top side of the interposer board are sharing their footprint with the locations for pin-head connectors from the bottom side. The following order of soldering such items is recommended:



Additionally, after soldering the middle pin-header row and before soldering the IC-socket, the exceeding pins on the top-side should be trimmed as far as possible with some side cutters. This will prevent the pins to collide with the IC-socket.

Additional Features Digit-Extension Interposer PCB

Some additional features of the interposer board are as follows:



- 1) Button to simulate collision of rocket with saucer missile: increases saucer score
- 2) Button to simulate collision of saucer with rocket missile: increases rocket score
- 3) 2 pin pin-head connector to supply 5VDC in case of underpowering (according to tests, not needed; in case of use, watch indicated polarity!)